

INTERNET OF THINGS

What is Internet of Things (IoT)?

Definition, history and features of IoT.

Nowadays, the internet-based information architecture allows the exchange of services and goods between all elements, equipment and objects connected to the network. The IoT refers to the networked interconnection of those everyday objects, which are often equipped with some kind of intelligence. In this context, Internet can be also a platform for devices to communicate electronically and share information and specific data with the world around them. So, IoT can be seen as a real evolution of what we know as Internet by adding more extensive interconnectivity, a better perception of the information and more comprehensive smart services. For the most part, the Internet was used for connection-oriented application protocols like **HTTP** (*Hypertext Transfer Protocol*) and **SMTP** (*Simple Mail Transfer Protocol*). However, nowadays a large number of smart devices communicate between themselves and to other control systems. This concept is known as **M2M** (*Machine-to-Machine communications*).

IoT (*Internet of things*) is an emerging global Internet-based technical architecture facilitating the exchange of goods and services in global supply chain networks has an impact on the security and privacy of the involved stakeholders.

Some highlights in the IoT history are the following:

- The term Internet of Things was first used by Kevin Ashton in 1999 that was working in the field of networked **RFID** (*radio frequency identification*) and emerging sensing technologies.
- However, IoT was “born” sometime between 2008 and 2009.
- In 2010, the number of everyday physical objects and devices connected to the Internet was around 12.5 billion. Nowadays there are about 25 billion of devices connected to the IoT. More or less a smart device per person [2].
- The number of smart devices or “things” connected to the IoT is expected to increase to a further 50 billion by 2020.

The IoT introduces a step change in individuals’ quality of life by offering a lot of new opportunities to data access, specific services in education, security, health care or transportation among others. On the other hand, it will be a key to increase enterprises’ productivity by offering a widely distributed, locally intelligent network of smart devices and new services that can be personalized to customer needs. The IoT brings benefits from improved management and tracking of assets and products, it increases the amount of information data and allows the optimization of equipment and use of resources that can be translated into costs saving. Moreover, it offers the opportunity to create new smart interconnected devices and explore new business models.

Applications

The number of applications and services that can provide IoT is practically unlimited and can be adapted to many fields of human activity by facilitating and enhancing their quality of life in multiple ways. This chapter gives a short list of applications and services based on IoT. However, it is just a limited description in order to understand all possible new applications and services that IoT could provide. An estimated value about \$19 trillion by 2020 is expected to be achieved by IoT applications and services.

IoT Applications and services:

- Connected intelligent buildings: Improvements in efficiency (energy management and saving) and security (sensors and alarms). Domotic applications including smart sensors and actuators to control home appliances. Health and education services at home. Remote control of treatments for patients. Cable/satellite services. Energy storage/generation systems. Automatic shutdown of electronics when not in use. Smart thermostats. Smoke detectors and alarms. Access control applications. Smart door locks. Sensors built into building infrastructure to guide first responders and assistances. Safety for all family members.
- Smart cities and transportation: Integration of security services. Optimization of public and private transportation. Parking Sensors. Smart management of parking services and traffic in real time. Smart management of traffic lights depending on traffic queues. Locate cars that have overstayed Smart energy grids. Security (cameras, smart sensors, information to citizens). Water management. Parks and Gardens irrigation. Smart garbage cans. Pollution and mobility controls. Get immediate feedback and opinions from citizens. Smart governance. Voting Systems. Accident monitoring, emergency actions coordination.



Fig. 1. Example of IoT applications: Smart cities.

- Education: Linking virtual and physical classrooms to make learning more efficient and accessible, e-learning. Access services to virtual libraries and educational portals. Interchange of reports and results in real time. Lifelong learning. Foreign languages learning. Attendance management.
- Consumer electronics: Smart phones. Smart TV. Laptops, computers and tablets. Smart refrigerators, washers and dryers. Smart home theatre systems. Smart appliances. Pet collar sensors. Personalization of the user experience.

Autonomous product operation. Personal locators. Smart glasses.

- Health: Monitoring of chronic diseases. Improvement of the quality of care and quality of life for patients. Activity Trackers. Remote diagnostic. Connected bracelets. Interactive belts. Sport and fitness monitoring. Intelligent tags for drugs. Drug usage tracking. Biochips. Brain-computer interfaces. Monitoring eating habits.
- Automotive: Smart Cars. Traffic control. Advance information about what is broken. Wireless monitoring of tire pressure of car. Smart energy management and control. Self-diagnosis. Accelerometers. Position, presence and proximity sensors. Analysis of the best way to go in real time. GPS tracking. Vehicle speed control. Autonomous vehicles using IoT services.
- Agriculture and environment: Measurement and monitoring of environmental pollution (CO₂, noise, contaminant elements presents in ambient). Forecasting climate changes based on smart sensors monitoring. Passive RFID tags attached to agriculture products. Sensors in pallets of products. Waste management. Nutrition calculations.
- Energy services: accurate data on energy consumption. Smart metering. Smart grids. Analysis and prediction of energy consumption behaviours and patterns. Forecasting future energy trends and needs. Wireless sensors networks. Energy harvesting and recycling.
- Smart Connectivity: Data management and service provisioning. Use of social media and social networking. Access to email, voice and video services. Interactive group communication. Real time streaming. Interactive gaming. Augmented reality. Network security monitoring. Wearable user interfaces. Affective computing. Biometric authentication methods. Consumer telematics. M2M communication services. Big data analysis. Virtual reality. Cloud computing services. Ubiquitous computing. Computer vision. Smart antennas.
- Manufacturing: Gas and flow sensors. Smart sensors of humidity, temperature, motion, force, load, leaks/levels. Machine vision. Acoustic and vibration sensing. Compound applications. Smart control of robots. Control and optimization of fabrication processes. Pattern recognition. Machine Learning. Predictive Analytics. Mobile logistics. Warehouse management. Prevent overproduction. Efficient logistics.
- Shopping intelligent shopping: RFID and other electronic tags and readers. Barcodes in retail. Inventory control. Control of geographical origin of food and products. Control food quality and safety.

Enabling technologies

Successful application of the IoT concept into the real world is possible thanks to advancements in underlying technologies. In this section the most relevant enabling technologies will be stated with the aim to provide a picture of the role they will likely play in the IoT.

4.1 Energy

Power and energy storage technologies are enablers for the deployment of IoT applications. Energy issues, in all its phases, from harvesting to conservation and usage, are central to the development of the IoT. These technologies have to provide high power-density energy generation and harvesting solutions which, when used with today's low power Nano electronics, will enable us to design self-powered intelligent sensor-based wireless identifiable device. There is still a need to research and develop solutions in this area (Nano electronics, semiconductor, sensor technology, micro systems integration) having as an objective ultra-low power devices, and more efficient and compact energy storage like batteries, fuel cells, and printed/polymer batteries, as current devices seem inadequate considering the processing power needed and energy limitations of the future. In addition, system integration will increase efficiency of current systems, and will provide a number of solutions for the future needs.

4.2 Sensors

Sensors are one of the key building blocks of the Internet of Things. As ubiquitous systems, they can be deployed everywhere. They can also be implanted under human skin, in a purse or on a T-shirt. Some can be as small as four millimeters in size, but the data they collect can be received hundreds of miles away. They complement human senses and have become indispensable in a large number of industries, from health care to construction. Sensors have the key advantage that they can anticipate human needs based on information collected about their context. Their intelligence multiplied by numerous networks allows them not only to report about external environment, but also to take action without human intervention.

Miniaturized silicon chips are designed with new capabilities in smaller form factors and better processing performance and efficiency. Costs are falling, following the Moore's Law. The cost of bandwidth has also declined and similarly the processing costs, enabling more devices to be not just connected, but smart enough to know what to do with all the new data they are generating or receiving.

Capabilities such as context awareness and inter-machine communication are considered a high priority for the IoT. Additional priorities are the integration of memory and processing power, the capacity of resisting harsh environments, and an affordable security. Furthermore, the development of ultra low power processors/microcontrollers cores designed specifically for mobile IoT devices and a new class of simple and affordable IoT-centric smart systems will be an enabling factor. The solutions in this respect will range from micro programmed finite state machines to the use of microcontrollers. The choice is a trade-off between flexibility, programmability, silicon area, and power consumption. The devices require some form of non-volatile storage (EEPROM/FRAM/Polymer), independent of whether this will be laser trimmed at the time of manufacture, one time programmable, or electrically rewritable. Rewritable non-volatile memory is clearly preferred for achieving high throughput during production test, and allows concurrently the benefit of user memory, programmability and storage of sensor data.

4.3 Cloud computing

Cloud computing is a model for on-demand access to a shared pool of configurable resources (e.g., computers, networks, servers, storage, applications, services, software) that can be provisioned as Infrastructure as a Service (IaaS) or Software

as a Service (SaaS). One of the most important outcomes of the IoT is an enormous amount of data generated from devices connected to the Internet [7]. Many IoT applications require massive data storage, huge processing speed to enable real time decision making, and high-speed broadband networks to stream data, audio, or video. Cloud computing provides an ideal back-end solution for handling huge data streams and processing them for the unprecedented number of IoT devices and humans in real time.

4.4 Communication

New, smart multi frequency band antennas, integrated on-chip and made of new materials are the communication means that will enable the devices to communicate. On-chip antennas must be optimized for size, cost and efficiency, and could come in various forms like coil on chip, printed antennas, embedded antennas, and multiple antenna using different substrates and 3D structures. Modulation schemes and transmission speed are also important issues to be tackled allowing multi-frequency energy efficient communication protocols and transmission rates. The communication protocols will be designed for Web oriented architectures of the IoT platform where all objects, wireless devices, cameras, PCs etc. are combined to analyze location, intent and even emotions over a network. New methods of effectively managing power consumption at different levels of the network design are needed, from network routing down to the architecture of individual devices.

4.5 Integration

Integration of smart devices into packaging, or better, into the products themselves will allow a significant cost saving and increase the Eco friendliness of products. The use of integration of chips and antennas into non-standard substrates like textiles and paper, and the development of new substrates, conducting paths and bonding materials adequate for harsh environments and for ecologically sound disposal will continue. System-in-Package (SiP) technology allows flexible and 3D integration of different elements such as antennas, sensors, active and passive components into the packaging, improving performance and reducing the tag cost. RFID inlays with a strap coupling structure are used to connect the integrated circuit chip and antenna in order to produce a variety of shapes and sizes of labels, instead of direct mounting.

4.6 Standards

IoT devices are quite diverse and measure different parameters and with different conventions and units of measure. Though competing proprietary protocols keep getting proposed, it is likely that open source standards will be one of the ways to get this data to interoperate.

Clearly, open standards are key enablers for the success of wireless communication technologies and, in general, for any kind of Machine-to-Machine communication. However, the need for faster setting of interoperable standards has been recognised an important element for IoT applications deployment. Clarification on the requirements for a unique global identification, naming and resolver is needed. Lack of convergence of the definition of common reference models, reference architecture for the Future Networks, Future Internet and IoT and integration of legacy systems and networks is a challenge that has to be addressed in the future.

Challenges

Reliability

Reliability aims to increase the success rate of IoT service delivery. It has a close relationship with availability as by reliability, we guarantee the availability of information and services over time. Reliability is even more critical and has more stringent requirements when it comes to the field of emergency response applications. In these systems, the critical part is the communication network which must be resilient to failures in order to realize reliable information distribution. Reliability must be implemented in software and hardware throughout all the IoT layers. In order to have an efficient IoT, the underlying communication must be reliable, because for example by an unreliable perception, data gathering, processing, and transmission can lead to long delays, loss of data, and eventually wrong decisions, which can lead to disastrous scenarios and can consequently make the IoT less dependable.

Reliability refers to the proper working of the system based on its specification.

Performance

Evaluating the performance of IoT services is a big challenge since it depends on the performance of many components as well as the performance of the underlying technologies. The IoT, like other systems, needs to continuously develop and improve its services to meet requirements of customers. The IoT devices need to be monitored and evaluated to provide the best possible performance at an affordable price for customers. Many metrics can be used to assess the performance of the IoT including the processing speed, communication speed, device form factor, and cost.

Performance evaluation of the individual underlying protocols and technologies, application layer protocols, and QoS have been reported in the literature, but the lack of a thorough performance evaluation for IoT applications is still an open issue.

Quality of service (QoS) is the overall performance of a telephony or computer network, particularly the performance seen by the users of the network.

Interoperability

End-to-end interoperability is another challenge for the IoT due to the need to handle a large number of heterogeneous things that belong to different platforms. Interoperability should be considered by both application developers and IoT device manufactures to ensure the delivery of services for all customers regardless of the specifications of the hardware platform that they use. For example, most of the smartphones nowadays support common communication technologies such as WiFi, NFC, and GSM to guarantee the interoperability in different scenarios. Also, programmers of the IoT should build their applications to allow for adding new functions without causing problems or losing functions while maintaining

integration with different communication technologies. Consequently, interoperability is a significant criterion in designing and building IoT services to meet requirements of customers. Beside variety of protocols, different interpretations of the same standard implemented by different parties presents a challenge for interoperability. To avoid such ambiguities, interoperability testing between different products in a test-bed like ETSI Plugtests would be helpful. PROBE-IT is a research project that aims to ensure the interoperability of validated IoT solutions that conducted interoperability tests like CoAP, 6LoWPAN, and IoT semantic interoperability.

It is a known fact that two different devices might not be interoperable, even if they are following the same standard. This is a major showstopper for wide adoption of IoT technologies. Future tags must integrate different communication standards and protocols that operate at different frequencies and allow different architectures, centralised or distributed, and be able to communicate with other networks unless global, well defined standards emerge.

Security and Privacy

Security presents a significant challenge for the IoT implementations due to the lack of common standard and architecture for the IoT security. In heterogeneous networks as in the case of the IoT, it is not easy to guarantee the security and privacy of users. The core functionality of the IoT is based on the exchange of information between billions or even trillions of Internet connection objects. One open problem in IoT security that has not been considered in the standards is the distribution of the keys amongst devices. On the other hand, privacy issues and profile access operations between IoT devices without interferences are extremely critical. Still, securing data exchanges is necessary to avoid losing or compromising privacy. The increased number of smart things around us with sensitive data necessitates a transparent and easy access control management in such a way that for example one vendor can just read the data while another is allowed to control the device. In this regard, some solutions have been proposed such as grouping embedded devices into virtual networks and only present desired devices within each virtual network. Another approach is to support access control in the application layer on a per-vendor basis.

Management

The connection of billions or trillions of smart devices presents service providers with daunting issues to manage the Fault, Configuration, Accounting, Performance and Security (FCAPS) aspects of these devices. This management effort necessitates the development of new light-weight management protocols to handle the potential management nightmare that will potentially stem from the deployment of the IoT in the coming years. Managing IoT devices and applications can be an effective factor for growing the IoT deployments. For example, monitoring the M2M communication of the IoT objects is important to ensure all times connectivity for providing on demand services. The Light-weight M2M (LWM2M) is a standard that is being developed by the Open Mobile Alliance to provide interface between M2M devices and M2M Servers to build an application agnostic scheme for the management of a variety of devices. It aims to provide M2M

applications with remote management capabilities of machine-to-machine devices, services, and applications. The NETCONF Light protocol is an Internet Engineering Task Force (IETF) effort for the management of constrained devices provides mechanisms to install, manipulate, and delete the configuration of network devices. It is capable of managing a broad range of devices from resource-constrained to resource-rich devices. The independently developed MASH IoT Platform is an example of a platform that facilitates the management (monitoring, control, and configuration) of IoT assets anywhere in real-time using an IoT dashboard on smartphones. Maintaining compatibility across the IoT layers also needs to be managed to enhance connectivity speed and to ensure service delivery. The Open Mobile Alliance (OMA) Device Management working group is specifying protocols and mechanisms for the management of mobile devices and services in resource constrained environments.

Manufacturing

Manufacturing challenges must be convincingly solved. Costs must be lowered to less than one cent per passive RFID tag, and production must reach extremely high volumes, while the whole production process must have a very limited impact on the environment, be based on strategies for reuse and recycling considering the overall life-cycle of digital devices and other products that might be tagged or sensor-enabled.

Future of IoT

It is possible to identify, for the years to come, four distinct macro-trends that will shape the future of internet technologies, together with the explosion of ubiquitous devices that constitute the future Internet of Things [9]:

1. The first one, sometimes referred as “exaflood” or “data deluge”, is the explosion of the amount of data collected and exchanged. As current networks are ill suited for this exponential traffic growth, there is a need by all the actors to re-think current networking and storage architectures. It will be imperative to find novel ways and mechanisms to find, fetch, and transmit data. One relevant reason for this data deluge is the explosion in the number of devices collecting and exchanging information as envisioned as the Internet of Things becomes a reality.

The term **exaflood**, coined by Bret Swanson of Progress&Freedom Foundation, refers to the growing torrent of data on the Internet.

2. The energy required to operate the intelligent devices will dramatically decrease. Already today many data centers have reached the maximum level of energy consumption and the acquisition of new devices has necessarily to follow the dismissal of old ones. Therefore, the second trend can be identified covering all devices and systems from the tiniest smart dust to the huge data centers: the search for a zero level of entropy where the device or system will have to harvest its own energy.
3. Miniaturization of devices is also taking place amazingly fast. The objective of a single-electron transistor is getting closer, which seems the ultimate limit, at least until new discoveries in physics.
4. Another important trend is towards autonomic resources. The ever growing complexity of systems will be unmanageable, and will hamper the creation of new services and applications, unless the systems will show self-* properties, such as self-management, self-healing and self-configuration.

As a general trend, as it becomes less expensive to integrate technology into physical objects, we will see more application and adoption of IoT. In consequence, IoT will have major implications for both business-to-business and business-to-consumer companies in the next years.

Fundamental challenges

- Connectivity: There will not be one connectivity standard that “wins” over the others. There will be a wide variety of wired and wireless standards as well as proprietary implementations used to connect the things in the IoT. The challenge is getting the connectivity standards to talk to one another with one common worldwide data currency.
- Power management: More things within the IoT will be battery powered or use energy harvesting to be more portable and self-sustaining. Line-powered equipment will need to be more energy efficient. The challenge is making it easy to add power management to these devices and equipment. Wireless charging will incorporate connectivity with charge management.
- Security: With the amount of data being sent within the IoT, security is a must. Built-in hardware security and use of existing connectivity security protocols is essential to secure the IoT. Another challenge is simply educating consumers to use the security that is integrated into their devices.
- Complexity: Manufacturers are looking to add connectivity to devices and equipment that has never been connected before to become part of the IoT. Ease of design and development is essential to get more things connected especially when typical RF programming is complex. Additionally, the average consumer needs to be able to set-up and use their devices without a technical background.
- Rapid evolution: The IoT is constantly changing and evolving. More devices are being added everyday and the industry is still in its naissance. The challenge facing the industry is the unknown. Unknown devices. Unknown applications. Unknown use cases. Given this, there needs to be flexibility in all facets of development. Processors and microcontrollers that range from 16– 1500 MHz to address the full spectrum of applications from a microcontroller (MCU) in a small, energy-harvested wireless sensor node to high-performance, multi-core processors for IoT infrastructure. A wide variety of wired and wireless connectivity technologies are needed to meet the various needs of the market. Last, a wide selection of sensors, mixed-signal and power-management technologies are required to provide the user interface to the IoT and energy-friendly designs.